PART I - CONSTRAINTS, VIEWS AND SQL

CHAPTER 7







OBJECTIVES:

- Define a database using SQL data definition language
- Write single table queries using SQL
- Establish referential integrity using SQL
- Write single and multiple table SQL queries
- Define and use three types of joins

WHAT IS SQL?

- SQL is a database language designed for the retrieval and management of data in a relational database.
- SQL is the standard language for database management.
- All the RDBMS systems like MySQL, MS Access, Oracle, Sybase, Postgres, and SQL Server use SQL as their standard database language.
- SQL programming language uses various commands for different operations. We will learn about the like DCL, TCL, DQL, DDL and DML commands in SQL with examples.

WHY USE SQL?

Here, are important reasons for using SQL

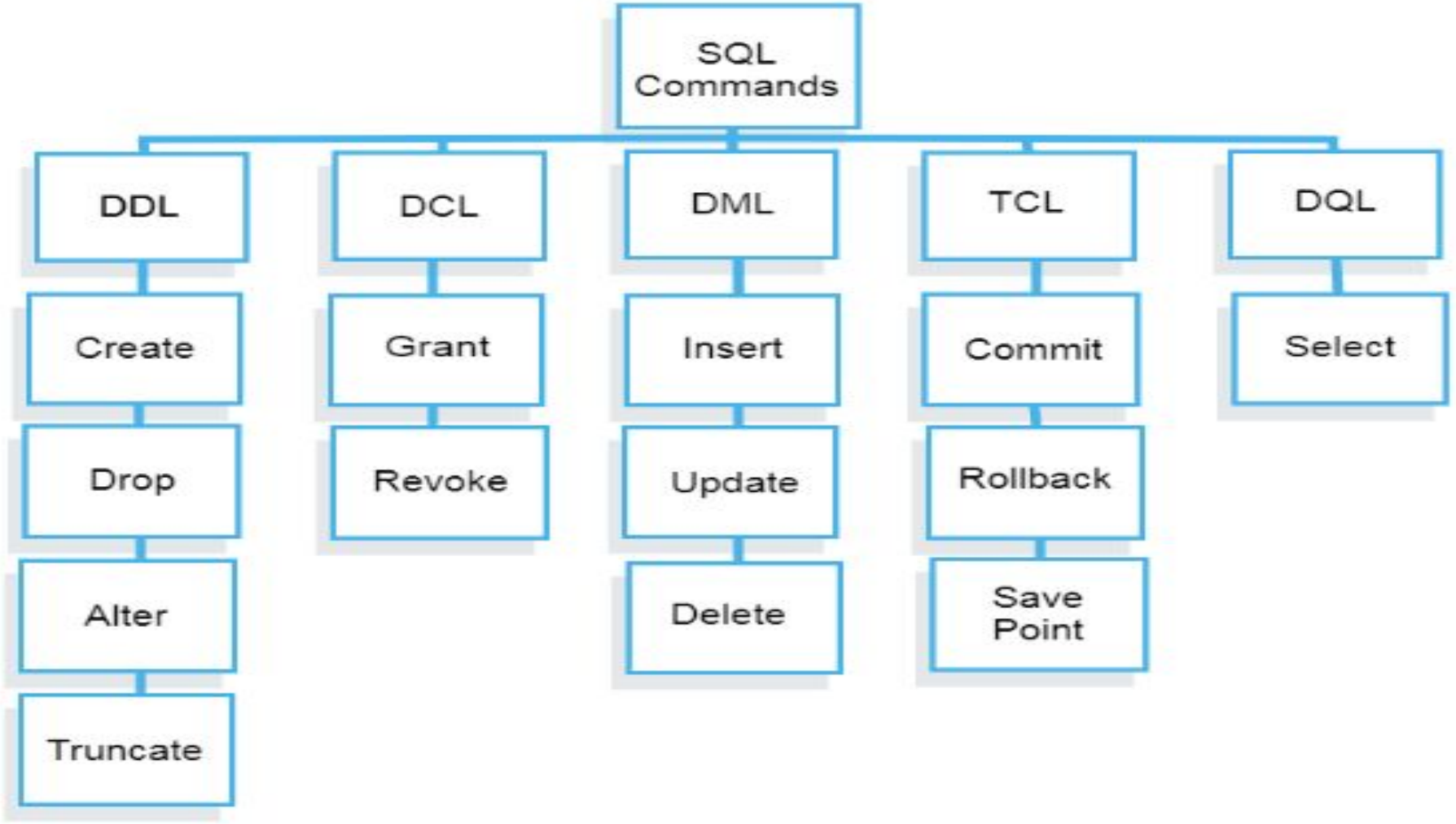
- It helps users to access data in the RDBMS system.
- It helps you to describe the data.
- It allows you to define the data in a database and manipulate that specific data.
- With the help of SQL commands in DBMS, you can create and drop databases and tables.
- SQL offers you to use the function in a database, create a view, and stored procedure.
- You can set permissions on tables, procedures, and views.

TYPES OF SQL

Here are five types of widely used SQL queries.

- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Data Control Language(DCL)
- Transaction Control Language(TCL)
- Data Query Language (DQL)

TYPES OF SQL


SQL Commands
DDL
DCL
DML
TCL
DQL
Create
Drop
Alter
Truncate
Grant
Revoke
Insert
Update
Delete
Commit
Rollback
Save Point
Select

WHAT IS DDL?

Data Definition Language helps you to define the database structure or schema. Let's learn about DDL commands with syntax.

Five types of DDL commands in SQL are:

CREATE

CREATE statements is used to define the database structure schema:

Syntax:

```
CREATE TABLE TABLE_NAME (COLUMN_NAME DATATYPES[,....]);
```

WHAT IS DDL?

For example:

```
Create database university;
Create table students;
Create view for_students;
```

WHAT IS DDL?

DROP

Drops commands remove tables and databases from RDBMS.

Syntax

```
DROP TABLE ;
```

For example:

```
Drop object_type object_name;
Drop database university;
Drop table student;
```

WHAT IS DDL?

ALTER

Alters command allows you to alter the structure of the database.

Syntax

To add a new column in the table

```
ALTER TABLE table_name ADD column_name COLUMN-definition;
```

To modify an existing column in the table:

```
ALTER TABLE MODIFY(COLUMN DEFINITION....);
```

For example:

```
Alter table guru99 add subject varchar;
```

WHAT IS DDL?

TRUNCATE

This command used to delete all the rows from the table and free the space containing the table.

Syntax:

```
TRUNCATE TABLE table_name;
```

For example:

```
TRUNCATE table students;
```